Craig McDaniel

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Summary

Software engineer with 20 years of development and leadership experience. Domain experience in finance, market research, health care information services, criminal justice, and distributed databases. Very passionate about data science, functional programming, building horizontally scalable applications, and generally improving the way we create software.

Specialties

Clojure, Go (Golang), Python, Ruby, Scala, Java, devops, data integration, rich user interfaces.

Experience

Datastax
Senior Software Engineer
Remote
January 2015 - present

Lifecycle Manager: A component of the OpsCenter product (monitoring and management of DSE/Cassandra clusters) responsible for provisioning and configuring DSE/Cassandra on clusters of host machines, VMs, or containers. I was one of three developers who started this greenfield project. One of the greatest achievements of this project was the creation of a "definition file" system capable of representing config file changes (new config properties, removed properties, template changes) across 20+ config files and across every public release of DSE/Cassandra. This enabled easy-to-use front-end form generation for configuring DSE - a daunting task without this feature. Technologies include: Clojure, Go (Golang), Python, and DSE/Cassandra configuration.

DSE/Cassandra Kubernetes Operator: Worked on the container team to develop Docker images and integration testing for DSE/Cassandra k8s operator. Technologies include: Kubernetes (k8s), Docker, Go (Golang), Clojure, and Gradle.

Infusion
Senior Consultant, Software Development
Raleigh, NC
May 2014 - December 2014

Harmony: As part of MetLife's "SWAT" team, participated in a major rewrite of a metadata-driven AngularJS application for the client. This program seeks to provide a common sales platform across their many regional businesses and insurance products. The system uses ontology to define the data in a way that can be configured to add new products and change the application forms without significant rewriting of the code. Ontologized data is mapped to and stored in a MongoDB database. The UI communicates via RESTful JSON calls to the backend (Ensemble).

Ensemble: The Java/Spring backend to Harmony. I was the designated engineer from my team to learn the "full stack". Utilized JAX-RS to provide RESTful web services. Integrated with external systems through a home-grown message bus called Symphony (built on top of JMS). Worked heavily on a configuration tool to generate forms and workflow data for the MongoDB based on data files managed by

the business.

MetLife Staffing: Conducted Tech 1 interview screenings for MetLife's staffing initiative in the Raleigh-Durham area. MetLife entrusted Infusion to screen the best candidates for these positions. Helped to make many subtle improvements to the Infusion process. Helped coach the technical recruiters to identify "weed-out" candidates before they get to Tech 1, saving the client money.

Clojure Evangelist: Unashamedly promoted Clojure within Infusion and MetLife SWAT. Gave presentations and ran "katas" for hands-on learning experience.

Deutsche Bank Global Technology
Assistant Vice President, Software Development
Cary, NC
April 2012 - May 2014

Horizon: Worked as a lead developer on the Horizon product, an internal, highly interactive executive dashboard and reporting tool. Using **GWT** and **HTML Canvas** with **XUL Runner** (stripped-down Firefox browser), Horizon made heavy use of animations, touch screen UI, and **bespoke data visualizations**. The Horizon team utilized a highly efficient **agile** process, with dedicated Product Owner, Functional Analyst, and QA roles. The Horizon platform used Google Web Toolkit (Java compiled down to JavaScript) on the client side. **Java**, **Spring framework**, **OSGI**, and **Oracle** were used on the backend.

Fandango: Took over as the sole developer and L3 support on Fandango, a data enrichment tool for Risk written in **Clojure**. Delivered a multi-regional version of Fandango well ahead of schedule and refactored the code to better support automated testing. Improved code coverage from non-existent to adequate. To date, *Fandango has had zero defects and no major incidents in production*. Fandango's stake holders continue to be amazed by its stability and performance.

Risk Access Point: Joined the RAP team with the goal to improve performance of their data access service. RAP provides a **REST**-based gateway (implemented in **Java**) to all of DB's Risk and Profit & Loss data, stored in the ODS (operational datastore) **Oracle** database. Improved performance by 60% in the first week since joining the team. Instrumental in revamping the team's process to make it more agile. Provided leadership and vision in designing the data discovery features of RAP.

Deutsche Asset Management
Assistant Vice President, Software Development
Louisville, KY
May 2008 – April 2012 (4 years)

Orchestra Platform: A suite of tools that facilitate investment research, product design, and the portfolio management and reporting requirements of DB's asset management businesses in the US, Europe, and Asia. Written in **Java** and uses the following frameworks and tools: **Adobe Flex** for the web front-end, **WebLogic** on the back-end, **Spring Framework** for dependency injection and configuration, **Eclipselink JPA** for Object Relational Mapping, **JMS** for "push" messaging to the Flex clients, **BIRT** and **Jasper** for reporting, **Quartz** for job scheduling, and **Oracle** for the database.

Planned and implemented three large-scale data migration projects to improve data quality and flexibility for new features. The Orchestra database makes heavy use of **temporal (historical) data patterns** and auditing.

Enterprise Data Integration: Consolidated data feeds from various sources for security ratings and analytics. Orchestra imports 20-30 million analytics per month. Used a federated approach to real-time price quotes for bonds and other fixed income securities - Barclay's and Bloomberg sources.

Implemented product blueprint filters using climate change ratings. This was to facilitate the launch of DB's innovative Climate Change Fixed Income strategy – one of the few of its kind. DB lunched this project July 1, 2010 with €15 million of seed capital.

Wrote custom Adobe Flex chart components to deliver beautiful interactive charts and graphs showing price history, signal changes, review levels, and episode capture.

Introduced **Scala** and **Clojure** languages into the project in areas with low risk, mostly unit testing and administrative capabilities.

Appriss
Senior Software Engineer, Core Team
Louisville, KY
July 2006 - May 2008 (1 year 11 months)

MethCheck: Lead developer for MethCheck Compliance Manager (electronic psuedoephedrine log). Written using **EJB3**. Used web services to provide universal access to any number of programmable client side devices (POS systems and embedded devices that include ID scanning, credit card scanning, signature scanning, and more). Over 6000 pharmacies nation-wide are using this system to ensure compliance with state laws and the federal Combat Meth Epidemic Act.

Developed an ID parser engine that decodes magnetic stripe, 1D, and 2D barcodes on the backs of licenses and other forms of ID. This component was used in MethCheck but is capable of being used standalone. Because very few states follow AAMVA standards, this project required in-the-field testing and debugging to continually improve quality.

Worked with a team to create a new, unified architecture for our data acquisition interfaces. The new system uses **Ruby on Rails** to allow rapid development of interface implementations and greatly simplifies support and management process.

Wrote a job scheduling manager in **Ruby** using the **ActiveRecord** framework.

Primary force behind Appriss adopting **Ruby** as a technology. Setup training for our entire software engineering department with Chad Fowler (well known in the Ruby community) as the trainer.

SHPS
Senior Software Engineer
Louisville, KY
May 2005 - July 2006 (1 year 3 months)

CareWise Navigator: Development and Engineering for CareWise Navigator Data Acquisition project. This product provides a shared data services hub that SHPS and third-party health care systems interface with. The solution uses **Hibernate** for ORM, a layer of **JMS** and **SOAP** services for client access, and **Spring Framework** to handle the configuration of component dependencies.

Helped to manage and "coach" a team of 7-8 offshore developers. Implemented extensive unit testing and best practice coding standards to improve quality from our offshore resources.

n+1, Inc.
Software Engineer II
Louisville, KY
September 2001 - May 2005 (3 years 9 months)

TraQline: Managed and extended this **Java** servlet-based marketing research web application being used by several Fortune 500 companies. TraQline is a survey of consumers and their purchases of consumer durable products. [http://www.traqline.com]

Shopper Survey: Project management and development of an **XML**-driven engine for creating web-based surveys, collecting survey data, and exporting the data for use in TraQline. This solution handles more than 30,000 respondents (over 2 million responses) during each one-week survey period.

Church Central: Worked with a team of engineers to build a **Java**-based content management system. [http://www.churchcentral.com]

Maintained several industry portal websites for Net World Alliance: [http://www.atmmarketplace.com, http://www.pizzamarketplace.com, http://www.kioskmarketplace.com]

Designed an **XML/XSL**-based mass-mailing system capable of delivering over 50,000 customized email

messages per hour.

Became a **certified instructor for SAIR Linux 101, LPI 101 and 102, and Java** programming training courses. Taught classes in Louisville, Frankfort, and Lexington.

Generated proposals, estimates, and scope documents for potential customers.

Miscellaneous system administration on over a dozen production Linux systems.

Internet Tool & Die
Software Engineer I
Louisville, KY
1998 - 2001 (3 years)

Developed web applications in **PHP, Python,** and **Java**.

Used and administered **MySQL**, **Solid**, and **PostgreSQL** databases.

Installed and administered Apache, Tomcat, and Zope web servers.

Assisted in the creation of time/cost estimates.

Skills & Expertise

Clojure, Ruby, Python, JavaScript, Scala, Groovy, Agile, Spring, JMS, XML, Web Services, JBoss, Glassfish, Weblogic, Oracle, MySQL, Ruby on Rails, Datomic, Hadoop.

Education

University of Louisville
Masters, Computer Engineering and Computer Science, 2000 - 2001

University of Louisville
Bachelor of Science, Engineering Math and Computer Science, 1995 - 2000

Highest Honors, Eagle Scout Trustee's Scholarship, Columbia HCA Scholarship, 1998 Kosair Charities Greek Scholar of the Year Award, 1998 Greek Men's Highest GPA Award, National Dean's List (1996 – 2001), Golden Key National Honor Society, Tau Beta Pi Engineering Honor Society, Speed Alumni Foundation Outstanding Senior Award, Student Linux Users Group (President), Kappa Sigma Fraternity (Treasurer, Rush Chairman), Association for Computing Machinery (ACM)

"Craig is a hard-working dedicated employee that will do whatever it takes to get the job done. He is the fastest learner and smartest person that I know."
— Matt Mitchell, MBA; Implementation Analyst; Appriss; worked with Craig at Appriss

"I came to Craig with what was, at the time, a very novel business concept for digital music distribution. In our first meeting, I was concerned because I rambled on for well over an hour in describing our logistical requirements and he only jotted down a few notes. Little did I know that he was able to encapsulate everything we needed in those notes and delivered a product above and beyond our expectations. Months later I still marveled at the forethought he put into his coding, making the system expandable for concepts we hadn't even thought of yet. Brilliant is an understatement."
— T. D. Ruth; was Craig's client at n+1, Inc.

"Craig has a solution for everything. He can go to any extent to find the solution. Very helpful and is ready to share his knowledge to his colleagues. He is exceptionally good at understanding any kind of open source in the Java community."
— Vineet Bhatt; Team Leader; Headstrong; worked directly with Craig at SHPS

“I had the pleasure of working with Craig on a high profile project. He was highly experienced, and an extremely productive polyglot lead engineer with impressive leadership skills. I would love the opportunity to work with him again.”
— Zeashan Pappa; Enterprise Solutions Architect; Infusion; worked with Craig at Infusion